

Assessing hypoxia risk and ecological stability in reservoir-based aquaculture in the Wonorejo reservoir, Indonesia

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Abstract: Sustainable aquaculture in inland reservoirs depends on seasonal hydrology, which influences nutrient dynamics and O₂ availability. This study aimed to quantify O₂ and nutrient-based ecological carrying capacity and seasonal hypoxia risk to support sustainable aquaculture planning. This study evaluated the feasibility and ecological capacity of aquaculture in the Wonorejo Reservoir, Indonesia, using a four-pillar framework: feasibility mapping, ecological assessment, carrying capacity estimation, and risk-based ecological management. Monthly water quality data from 2022–2024, supported by *in situ* validation in March 2025, were analysed using the Vollenweider phosphorus model, O₂ balance calculations, total oxygen demand (TOD), and the ecological balance index (EBI). Spatial analysis identified areas suitable for aquaculture in about 36.2% of the area. Phosphorus loading remained within oligotrophic conditions (0.12–0.28 mg P·m⁻²·y⁻¹), indicating that nutrient enrichment is not the dominant limiting factor. Oxygen availability emerged as the dominant ecological constraint under the applied O₂ balance framework. Dry season elevated TOD relative to estimated natural re-aeration capacity. This resulted in chronic O₂ deficits (EBI = 0.50–0.60) and reduced the system's resilience. Under these O₂-limited conditions, theoretical N-based aquaculture potential (1,250–1,600 mg·y⁻¹) declined to a sustainable range of 500–900 mg·y⁻¹. A hypoxia-focused hazard–exposure–vulnerability (HEV) assessment showed high HEV during dry seasons, moderate risk during transitional periods, and low risk during rainy conditions. The findings emphasise that O₂ dynamics, rather than nutrient levels, govern sustainable aquaculture capacity. Oxygen-based stocking regulation, organic load management, aeration, and spatial zoning are recommended to improve ecological stability.

Keywords: ecological carrying capacity, oxygen balance, stocking density, sustainable aquaculture, trophic status

INTRODUCTION

Aquaculture is one of the fastest-growing food production sectors globally, contributing substantially to food security and rural livelihoods (Garlock *et al.*, 2022). In tropical regions, floating net cage aquaculture (FNC) in reservoirs has expanded rapidly, particularly for Nile tilapia (*Oreochromis niloticus*), due to its rapid growth and environmental tolerance (Amthunghong *et al.*, 2025). However, increasing production intensity introduces organic matter and nutrient inputs, potentially altering O₂ fluctuations, nutrient cycling, and overall ecosystem stability (Okechi *et al.*, 2025).

From a classical limnological perspective, dissolved oxygen (DO) represents a dynamic balance between O₂ production (photosynthesis and atmospheric reaeration) and O₂ consumption (community respiration, organic matter decomposition, nitrification, and sediment O₂ demand) (Stumm and Morgan, 1996; Wetzel, 2001). Hypoxia develops when O₂ demand exceeds supply, particularly under prolonged residence time, reduced inflow, thermal stratification, or elevated organic loading (Baxa *et al.*, 2021; Yan *et al.*, 2021). Reservoir morphometry, hydrological variability, and mixing regime strongly regulate this balance (Szarek-Gwiazda, Pocięcha and Wilk-Woźniak, 2025). Even in systems with relatively low external phosphorus loading, O₂ depletion may arise if metabolic demand accumulates faster than physical renewal processes, especially during dry seasons characterised by limited flushing (Sun *et al.*, 2022).

Classical eutrophication theory emphasises phosphorus as the principal driver of algal biomass and subsequent O₂ depletion (Chorus and Spijkerman, 2021). However, contemporary studies demonstrate that O₂ limitation may also emerge from internal processes and organic waste accumulation associated with aquaculture activities, even under oligotrophic phosphorus conditions (Delmiro Rocha and Lima Neto, 2022). Therefore, ecological carrying capacity in reservoirs must consider not only nutrient thresholds but also the system's metabolic capacity to assimilate organic loads while maintaining sufficient O₂ availability (Zhou *et al.*, 2024).

Ecological carrying capacity (EEC) is the maximum biomass that causes unacceptable deterioration in ecosystem functioning (Bojarski *et al.*, 2022). In aquaculture systems, EEC is often estimated using nutrient-based mass-balance approaches; however, oxygen-based constraints may be a more immediate ecological bottleneck, particularly in tropical reservoirs where high temperatures increase respiration rate and O₂ demand (Barnes, King and Carter, 2011). Many applied assessments remain nutrient-oriented and insufficiently grounded in process-based aerobic patterns. In Indonesia, many reservoirs serve as multifunctional water infrastructure rather than dedicated aquaculture zones (Sholichin *et al.*, 2024a). The Wonorejo Reservoir in East Java exemplifies this system, with primary management objectives including irrigation supply, flood control, hydropower generation, raw water provision, and tourism. Unlike intensively farmed reservoirs with floating net cages, Wonorejo is not designated for commercial cage aquaculture but is managed for capture fisheries and ecosystem conservation. Fish populations are maintained through periodic restocking (e.g., tilapia and carp) to support local fisheries and recreational angling.

This study addresses the research question of whether seasonal hypoxia risk constrains ecological stability and the

potential for aquaculture in a multifunctional tropical reservoir. Specifically, we examine the hypothesis that O₂ availability, rather than phosphorus loading alone, represents the dominant ecological limitation influencing sustainable aquaculture development in the Wonorejo Reservoir. To test this, classical limnological principles of O₂ balance and nutrient mass loading are applied within a carrying capacity framework to evaluate seasonal hypoxia risk and its implications for ecological stability. By comparing O₂-based and nutrient-based constraints under prevailing hydrological conditions, the study aims to determine the primary regulating factor of sustainable reservoir-based aquaculture.

MATERIALS AND METHODS

STUDY AREA DESCRIPTION

The study was conducted in the Wonorejo Reservoir, located on the slopes of Mount Wilis in Wonorejo Village, Pagerwojo District, Tulungagung Regency, East Java, Indonesia. The reservoir lies 12–15 km northwest of Tulungagung city centre at 8°00'58.61" S and 111°47'58.34" E (Fig. 1). Construction began in 1994 and finished in 2001. Covering 3.85 km², the reservoir receives inflows from the Mbodeng and Kali Gondang Rivers (Amelia, Wahyuni and Harisuseno, 2021). With a discharge of 820 m³·s⁻¹ and a storage volume of 109.11 mln m³, it supports water supply, irrigation, hydropower, tourism, capture fisheries, and floating net cage (FNC) aquaculture. The surrounding area includes plantations, rice fields, and settlements. Air temperatures range between 25 and 29°C, with 11.6 km·h⁻¹ wind speeds and rainfall up to 288 mm (January). These factors influence water mixing, sediment resuspension, and nutrient transport.

WATER QUALITY DATA AND SAMPLING

Water quality data (2022–2024) were obtained from Jasa Tirta I, a state-owned Indonesian water resources management agency responsible for reservoir monitoring. Monthly measurements at Station 2 included temperature, pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), ammonia, nitrate, nitrite, and phosphate, following standardised protocols aligned with APHA, AWWA, and WEF (2017). Field validation was conducted by the authors in March 2025 at three locations (inlet, mid-reservoir, and dam). The DO and pH were measured *in situ* using a calibrated multiparameter probe, while nutrient and organic samples (0.5 m of depth) were analysed within 24 h according to APHA, AWWA, and WEF (2017). Agreement between datasets was strong (root mean square error (RMSE) < 10%), supporting data reliability.

Suitability was evaluated using the weighted arithmetic water quality index (WAWQI) (Lukhabi *et al.*, 2023), with parameters scaled to a 1–5 tilapia suitability range based on ecological importance (Tab. 1). It is important to distinguish ecological status indices from aquaculture suitability indices. While ecological assessments prioritise minimal nutrient concentrations to prevent eutrophication, aquaculture requires moderate nutrient levels to support primary productivity. Therefore, thresholds and weights in this study were defined based on optimal tilapia production conditions rather than pristine limnological standards. Spatial suitability analysis was conducted

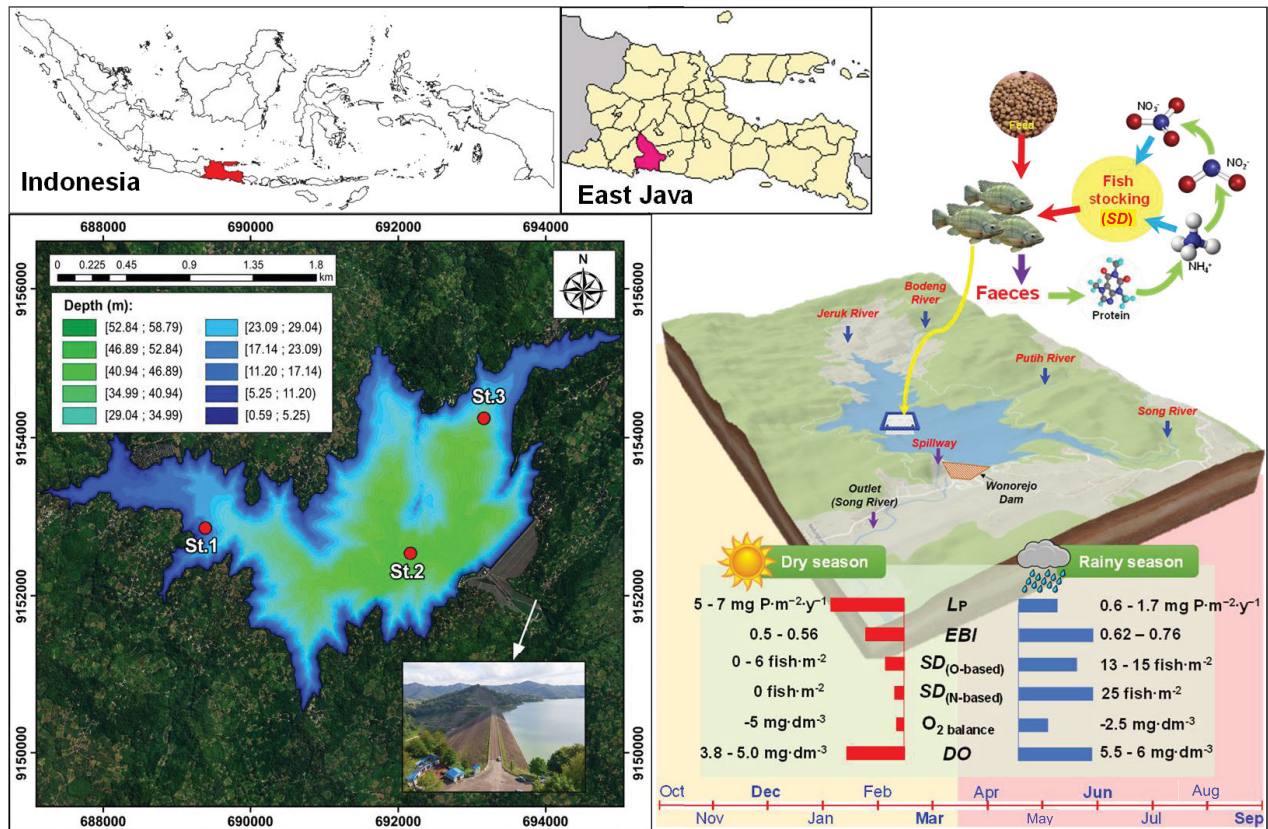


Fig. 1. Wonorejo Reservoir, Indonesia; St. = station, L_p = phosphorus loading, EBI = ecological balance index, SD = stocking density, DO = dissolved oxygen; source: own elaboration

Table 1. Scoring matrix of water quality suitability for Nile tilapia aquaculture

Parameter, weight	Value in suitability class					Reference
	5	4	3	2	1	
	highly suitable	suitable	moderately suitable	less suitable	unsuitable	
Temperature ($^{\circ}\text{C}$), 15%	27–32	23–27	18–22	12–18	<10 and >28	Hamed <i>et al.</i> (2021)
DO ($\text{mg}\cdot\text{dm}^{-3}$), 12.5%	>7	4–7	3.5–4	3	<3	Jiménez <i>et al.</i> (2024)
pH, 15%	6.5–9	7–8	6.6–7	5–6.6	<5 and >9	Wu <i>et al.</i> (2018)
NH_3 ($\text{mg}\cdot\text{dm}^{-3}$), 12.5%	<0.02	0.02–0.05	0.05–0.1	0.1–0.2	>0.2	Xu <i>et al.</i> (2021)
PO_4 ($\text{mg}\cdot\text{dm}^{-3}$), 10%	>0.201	0.101–0.2	0.051–0.100	0.021–0.05	0–0.02	David <i>et al.</i> (2022)
NO_2 ($\text{mg}\cdot\text{dm}^{-3}$), 5%	0–0.001	0.001–0.02	0.02–0.03	0.02–0.05	>0.05	Zhang <i>et al.</i> (2023)
NO_3 ($\text{mg}\cdot\text{dm}^{-3}$), 5%	1–5	5–10	10–20	20–50	>50	Zhang <i>et al.</i> (2023)
Depth (m), 3%	7–20	3–6	3	2–3	<2 and >20	Okechi <i>et al.</i> (2025)
Transparency (cm), 2%	30–40	10–20	20–30	>40	<10	Okechi <i>et al.</i> (2025)
COD ($\text{mg}\cdot\text{dm}^{-3}$), 10%	<10	11–20	20–30	30–40	>40	Moyo and Rapatsa (2021)
BOD ($\text{mg}\cdot\text{dm}^{-3}$), 10%	<3	3–10	10–20	20–25	>25	Xu and Boyd (2016)

Explanations: DO = dissolved oxygen, COD = chemical oxygen demand, BOD = biochemical oxygen demand.

Source: own elaboration based on literature.

in ArcGIS 10.8 using zonation based on observed station gradients. Due to the limited number of sampling stations, spatial visualisation was performed as a deterministic classification rather than formal geostatistical kriging. The interpolated surfaces were reclassified into five suitability classes and

integrated with bathymetry and land-use layers. The resulting WQI map identified S1 (highly suitable), S2 (suitable), and N (unsuitable) zones. Spatial outputs should therefore be interpreted as management-oriented zonation rather than high-resolution predictive interpolation.

WATER QUALITY ASSESSMENT AND ESTIMATION METHODOLOGY

The analytical procedure for assessing the hypoxia follows the framework in Figure 2.

1. Nutrient-based carrying capacity. Phosphorus loading was estimated using the Vollenweider model, with residence time from the inflow-outflow balance. Areal P-loading values were used to classify trophic status.
2. Total oxygen demand (*TOD*). The *TOD* was calculated from biochemical oxygen demand (*BOD*), chemical oxygen demand (*COD*), NH_3 , and NO_2 , with *COD* adjusted to represent only the biodegradable fraction.
3. Ecological balance index (*EBI*). The dissolved oxygen (*DO*), $\text{PO}_4\text{-P}$, and *BOD* were normalised into ecological scores and combined using a weighted average to indicate ecosystem balance and stress levels.
4. O_2 -based carrying capacity. The surplus of O_2 (*DO* minus ecosystem demand) was converted into an O_2 . This was used to estimate potential harvest and derive stocking density based on target fish size.
5. N-based carrying capacity. Total nitrogen (*TN*) was calculated from NO_3 , NO_2 , and NH_3 , and N load was estimated from feed inputs to determine allowable stocking density within N thresholds.
6. Hypoxia risk assessment. The hazard-exposure-vulnerability (*HEV*) model was used to assess hypoxia vulnerability, with biomass risk and production loss derived from the gap between O_2 - and N-supported biomass.

RESULTS

WATER QUALITY

Water quality in the Wonorejo Reservoir during 2022–2024 remained suitable for *O. niloticus* culture, with most parameters within biological limits despite seasonal variation. Temperature

(25–30°C) and pH (6.9–8.45) consistently supported tilapia metabolism and nitrification. Dissolved oxygen (*DO*) ($3.8\text{--}6.0\text{ mg}\cdot\text{dm}^{-3}$) declined in June–July 2024 due to decomposition and reduced reaeration, but generally remained above the $3.7\text{ mg}\cdot\text{dm}^{-3}$ stress threshold. Although no acute hypoxia ($<2\text{ mg}\cdot\text{dm}^{-3}$) occurred, *DO* frequently approached the $5\text{ mg}\cdot\text{dm}^{-3}$ ecological threshold, indicating seasonal reductions in O_2 buffering capacity rather than severe hypoxic events. Ammonia ($0.049\text{--}0.300\text{ mg}\cdot\text{dm}^{-3}$) remained below the toxic limit ($<0.8\text{ mg}\cdot\text{dm}^{-3}$), nitrate ($<0.0321\text{--}3.737\text{ mg}\cdot\text{dm}^{-3}$) stayed far below the $20\text{ mg}\cdot\text{dm}^{-3}$ tolerance limit (Rahardjo, Shih and Fan, 2024), and nitrite ($<0.0007\text{--}0.061\text{ mg}\cdot\text{dm}^{-3}$) remained well under the $0.3\text{ mg}\cdot\text{dm}^{-3}$ threshold, indicating stable nutrient dynamics.

Organic load indicators fluctuated moderately, with biochemical oxygen demand (*BOD*) ($4.3\text{--}7.19\text{ mg}\cdot\text{dm}^{-3}$) slightly exceeding the optimal $4\text{--}6\text{ mg}\cdot\text{dm}^{-3}$ range at times (Gymnastiar *et al.*, 2025), and chemical oxygen demand (*COD*) ($18.3\text{--}29.86\text{ mg}\cdot\text{dm}^{-3}$) showing occasional spikes from runoff but remaining manageable. Phosphate levels ($<0.0089\text{--}0.093\text{ mg}\cdot\text{dm}^{-3}$) reflected oligotrophic conditions, while reservoir depth ($0.69\text{--}58.79\text{ m}$) influenced circulation and stratification patterns (Rahardjo *et al.*, 2026). Overall, despite seasonal shifts, water quality remained favourable for sustainable tilapia aquaculture.

OVERALL SUITABILITY OF THE WONOREJO RESERVOIR

The water quality parameter data from the Wonorejo Reservoir, collected during observations in 2022, 2023, and 2024, were validated against primary data obtained through direct measurements in 2025 using the root mean square error (*RMSE*) method (Supplementary file S1). The results showed that the overall deviation remained below 10% and was highly accurate in assessing the hypoxia in the Wonorejo Reservoir. The integrated water quality assessment (Supplementary file S2) revealed that the Wonorejo Reservoir exhibited mean suitability scores of 72.5%, 77.0%, and 78.5% in 2022, 2023, and 2024, respectively. Based on a classification, the reservoir is categorised as “suitable” for

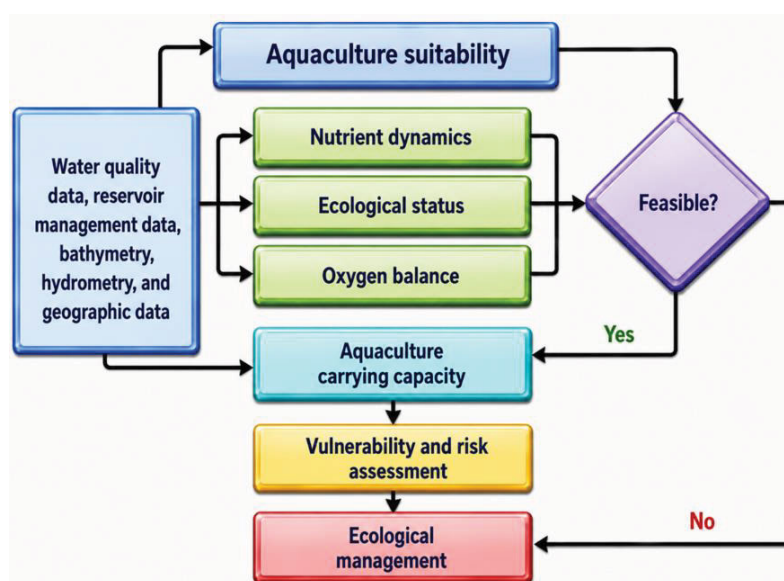


Fig. 2. Conceptual framework for the assessment of hypoxia in the reservoir for sustainable aquaculture; source: own elaboration

O. niloticus culture using floating net cage aquaculture (FNCs). Despite overall favourable conditions, several parameters, particularly phosphate, ammonia, and COD, deviated from optimal levels, indicating localised ecological stress (Sholichin *et al.*, 2024b).

SPATIAL DISTRIBUTION ANALYSIS OF AQUACULTURE SUITABILITY PARAMETERS

Spatial analysis of the Wonorejo Reservoir showed spatial variations influencing the suitability of FNCs culture for *O. niloticus*. Temperature and pH were uniform and consistently within optimal ranges, indicating stable thermal and buffering conditions. In contrast, high transparency rendered many areas unsuitable due to excessive light penetration that may increase thermal instability and promote algal fluctuations. The parameters related to O₂ displayed moderate gradients: DO was generally adequate but declined toward deeper, less-circulated zones, while BOD and COD reflected moderate organic enrichment in mid-reservoir areas. Nutrient concentrations were relatively low and spatially stable; ammonia, nitrate, and nitrite remained below toxicity limits, and low phosphate indicated oligotrophic conditions that reduce eutrophication risk. Bathymetric variation also shaped suitability by providing more favourable circulation and light conditions in shallower zones. Overall, integrated spatial assessment showed moderate to high suitability, with downstream areas performing best and an average suitability index of 77%, covering about 36.2% of the reservoir (Fig. 3).

DISCUSSION

NUTRIENT DYNAMICS: STABLE BUT SEASONALLY VULNERABLE

Phosphorus loading (L_P) in the Wonorejo Reservoir was quantified using the Vollenweider model, which estimates critical areal loading ($L_P = \frac{P \cdot z \cdot \rho}{1 + \sqrt{tw}}$, g P·m⁻²·y⁻¹) based on mean phosphorus concentration (P), reservoir depth (z), flushing/settling

factor (ρ), and water residence time ($tw = \frac{(365 \cdot 24 \cdot 3600) \cdot (Q_{in} - Q_{out})}{ECS}$). The water residence time (tw) was calculated from inflow (Q_{in}), outflow (Q_{out}), and the reservoir's effective capacity storage (ECS). The resulting L_P values were interpreted using standard trophic categories, from oligotrophic (<0.5 g P·m⁻²·y⁻¹) to hypereutrophic (>2.0 g P·m⁻²·y⁻¹). Model inputs were derived from bathymetric data ($z \approx 24$ m), phosphorus concentrations, and seasonal hydrological records. The flushing factor (ρ) was set to 1. Inflow–outflow values used to compute tw were: December ($Q_{in} = 8.77$ – 9.48 m³·s⁻¹; $Q_{out} = 5$), March (8.64–12.44; 6), June (1.02–1.73; 5), and September (0; 5).

Seasonal L_P patterns revealed that dry-season values increased to 5.6–7.2 mg P·y⁻¹ due to reduced inflows, longer residence time, and accumulation of organic particulates from aquaculture and catchment sources (Yin *et al.*, 2022). During the rainy season, L_P declined to 0.62–2.86 mg P·y⁻¹, reflecting enhanced flushing, dilution, and sedimentation (Delmiro Rocha and Lima Neto, 2022). Despite these fluctuations, all L_P values remained within oligotrophic limits (Fig. 4a), indicating low eutrophication risk under current loading conditions. However, the elevated dry-season L_P underscores a seasonal vulnerability to nutrient buildup, highlighting the need for strengthened nutrient control and adaptive aquaculture management to prevent future eutrophication (Sun *et al.*, 2022).

THE O₂ BALANCE – THE DECISIVE BOTTLENECK

Total oxygen demand (TOD), the combined O₂ consumption from decomposition and nitrification, was calculated using this equation: $TOD = BOD + 0.1 COD + 4.57 NH_3 + 3.43 NO_2$, where chemical oxygen demand (COD) was scaled to reflect the biodegradable fraction. Using water quality data, TOD and seasonal O₂ deficits were estimated (Fig. 4b). The TOD ranged from 7.41 to 9.65 mg·dm⁻³, exceeding available dissolved oxygen (DO) (3.8–6.0 mg·dm⁻³) and producing persistent O₂ deficits (from –2.24 to –5.23 mg·dm⁻³), confirming that O₂ consumption surpassed natural reaeration and photosynthesis (Yan *et al.*, 2021).

Seasonality intensified these imbalances. During the dry season, low inflows and longer residence times increased organic

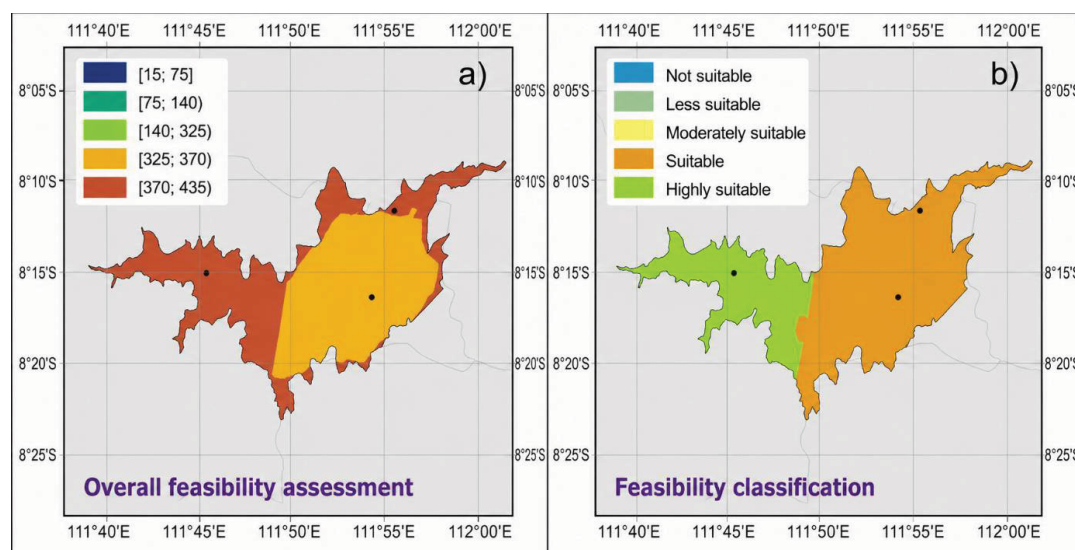


Fig. 3. Aquaculture of Wonorejo Reservoir: a) spatial distribution, b) suitability; source: own study

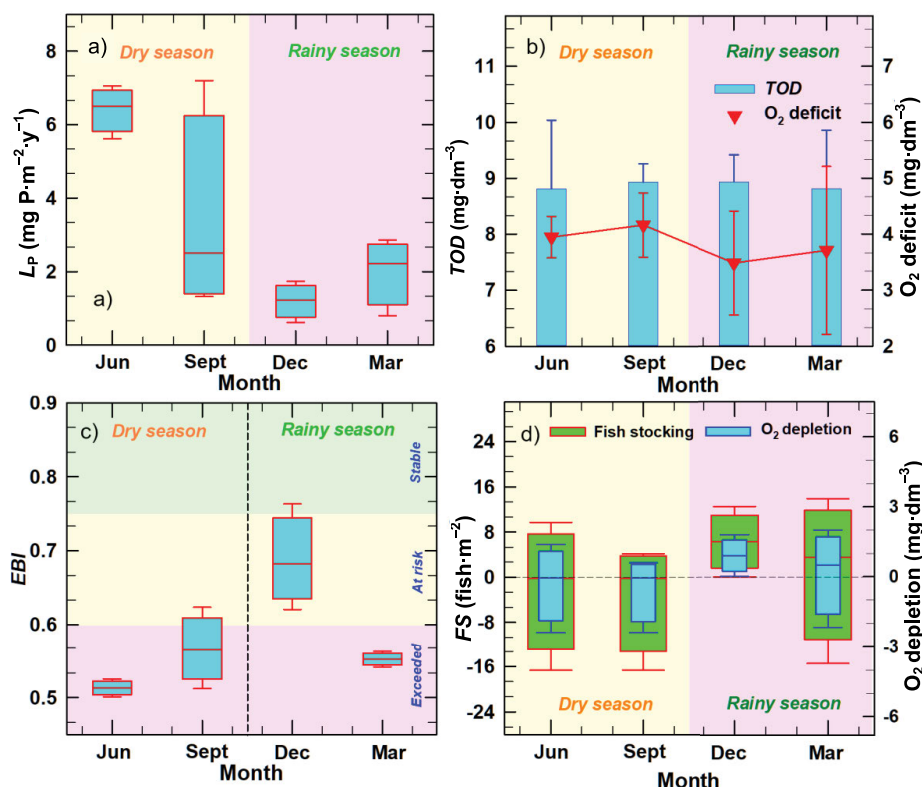


Fig. 4. Model-derived seasonal outputs for the Wonorejo Reservoir: a) phosphorus loading (L_p), b) O_2 balance carrying capacity, c) ecological balance index (EBI), d) integrated aquaculture carrying capacity; TOD = total oxygen demand, FS = fish stocking; values are deterministic calculations based on seasonal-mean input parameters; source: own study

accumulation, resulting in the lowest DO ($3.8 \text{ mg}\cdot\text{dm}^{-3}$) and O_2 deficit ($-5.23 \text{ mg}\cdot\text{dm}^{-3}$) relative to the $5 \text{ mg}\cdot\text{dm}^{-3}$ precautionary ecological threshold. In contrast, rainy-season mixing enhanced O_2 renewal, raising DO to $\sim 5.9 \text{ mg}\cdot\text{dm}^{-3}$ and reducing deficits. These patterns indicate that O_2 availability is the primary carrying-capacity constraint within the precautionary management framework, despite the absence of acute hypoxia ($< 2 \text{ mg}\cdot\text{dm}^{-3}$) or observed fish mortality. The results, therefore, reflect reduced seasonal O_2 -buffering capacity rather than crisis-level depletion. Accordingly, adaptive measures such as improved feed management, optimised aeration, and alignment of culture cycles with hydrological conditions are necessary to maintain sustainable production margins under seasonal variability.

ECOLOGICAL STATUS AND CHRONIC STRESS

The ecological balance index (EBI) provides an integrated assessment of ecology by combining three key indicators: DO , $PO_4\text{-P}$, and biochemical oxygen demand (BOD). Nutrient status is represented by $PO_4\text{-P}$, with oligotrophic conditions ($< 0.05 \text{ mg}\cdot\text{dm}^{-3}$) scored as 1.0, mesotrophic ($0.05\text{--}0.15 \text{ mg}\cdot\text{dm}^{-3}$) as 0.7, and eutrophic ($> 0.15 \text{ mg}\cdot\text{dm}^{-3}$) as 0.4. Organic loading is evaluated using BOD ($BOD_{\text{score}} = 1 - \frac{BOD-3}{7}$), while DO performance is normalised via $DO_{\text{score}} = \frac{DO}{8}$. These components are then integrated using a weighted formulation ($EBI = (0.4 DO_{\text{score}}) + (0.4 PO_{4\text{score}}) + (0.2 BOD_{\text{score}})$), where DO and $PO_4\text{-P}$ each contribute 40% and BOD contributes 20%. The EBI is therefore a process-informed tool for comparative seasonal assessment rather than a standardised ecological integrity index.

Greater weight was assigned to DO and $PO_4\text{-P}$ due to their primary influence on O_2 balance, while BOD was weighted lower as a secondary indicator. The EBI scores are interpreted as stable (≥ 0.8), at risk ($0.6\text{--}0.8$), or ecological stress (< 0.6).

Applying this framework to the Wonorejo Reservoir (Supplementary File S6) shows that EBI values ranged from 0.50 to 0.76 (Fig. 4c), remaining below the stability threshold. The lowest dry-season values (0.50–0.56) reflect reduced inflow and prolonged residence time, which increased organic accumulation and respiration pressure, lowering DO performance within the index. Rainy-season mixing improved dilution and O_2 renewal, raising EBI to 0.62–0.76; however, recovery remained incomplete as nutrient inputs and subsequent respiration continued to constrain O_2 buffering capacity. Thus, seasonal EBI dynamics reflect the interactions among O_2 supply, organic decomposition, and productivity processes under hydrological variability (Baxa *et al.*, 2021).

AQUACULTURE POTENTIAL AND SEASONAL CONSTRAINTS

Aquaculture carrying capacity in the Wonorejo Reservoir was quantified using an O_2 balance that accounts for seasonal DO variability. The O_2 surplus ($O_{2\text{Sur}} = DO - O_{2\text{eco}}$), the remaining DO after satisfying ecological O_2 demand ($O_{2\text{eco}}$), represents the O_2 available to support fish biomass and is converted into total O_2 deposit ($O_{2\text{dep}} = O_{2\text{Sur}} \cdot A \cdot D \cdot 1000$) based on cage area, depth, and a conversion factor of 1000. The $O_{2\text{eco}}$ $5 \text{ mg}\cdot\text{dm}^{-3}$ was applied as a precautionary ecological protection threshold to prevent hypoxic stress and maintain reservoir stability. The DO

5 mg·dm⁻³ is reserved for ecosystem functioning, and only positive O_{2Sur} is allocated for aquaculture carrying capacity and conservative sustainability-based management rather than a dynamic physiological requirement. This O₂ pool is then used to estimate potential harvest ($F_h = \frac{O_{2dep}}{FOD \cdot Ht}$) from fish oxygen demand (FOD) and culture duration (Ht), with stocking density ($SD = \frac{F_h}{Z_h}$) derived using the number of fish per kg at harvest (Z_h). Using DO data, an O_{2eco} of 5 mg·dm⁻³, tilapia FOD of 200 mg O₂ kg⁻¹·h⁻¹ (≈ 4.8 mg O₂ kg⁻¹·d⁻¹), a 120-day culture period, a harvest weight of 250 g (4 fish·kg⁻¹), and a 1 m² × 2 m floating net cage (FNC) configuration, O_{2Sur} and carrying capacity were quantified (Tab. 2, Fig. 4d).

Results show that O₂ supply-and-demand mechanisms impose strong seasonal constraints on aquaculture potential. The O₂ surplus ranged from -2.4 to 2.0 mg·dm⁻³, yielding stocking capacities from -17 to 14 fish·m⁻². Negative stocking densities represent mathematical artifacts arising from O₂ deficits and were therefore truncated to zero, indicating no sustainable additional stocking under those conditions. Rainy-season DO elevations generated positive O₂ balances and enabled moderate stocking (e.g., 0.8–2.0 mg·dm⁻³ yielding 6–14 fish·m⁻²), while dry-season hypoxia (-2.4 mg·dm⁻³) reduced capacity to zero. These findings indicate that O₂ availability functions as the primary carrying-capacity constraint within the precautionary framework, rather than evidence of acute hypoxia or observed fish mortality (Barnes, King and Carter, 2011). Therefore, dry-season aquaculture should be minimised or aeration-supported, whereas the rainy season permits limited intensification.

N-BASED CARRYING CAPACITY: SEASONAL NUTRIENT VULNERABILITY

The N-based carrying capacity was evaluated by quantifying total nitrogen (TN) and determining the nitrogen load (Nf) and N-based stocking density (SD_N). The TN was calculated as the sum of nitrate, nitrite, and ammonia ($[TN] = [NO_3^-] + [NO_2^-] + [NH_3]$), while nitrogen from fish ($Nf = \frac{Hw \cdot FR \cdot Pf \cdot Ncf}{Ht \cdot V \cdot 1000}$) incorporated harvest weight (Hw), feeding rate (FR), feed protein content (Pf), N conversion factor (Ncf), culture duration (Ht), and reservoir volume (V). The allowable N increment was defined as $\Delta N = 0.3 \text{ mg} \cdot \text{dm}^{-3} - TN$, based on the oligotrophic limit, and the maximum number of fish ($SD_N = \frac{\Delta N}{Nf \cdot A \cdot z}$), which integrates ΔN with N load, cage area (A), and mean depth (z). The TN represents dissolved inorganic N (DIN), so the N-based carrying capacity provides an operational measure of water-column N pressure rather than a full ecosystem N budget (complete N mass balance; N-organic, speciation dynamics, and sedimentation were not explicitly modelled).

Wonorejo Reservoir parameters ($FCR = 1.0$, $Pf = 30\%$, $FR = 5\% \text{ BW} \cdot \text{d}^{-1}$, $Ncf = 16\%$, $Hw = 250 \text{ g}$, $Ht = 120 \text{ days}$, $V = 10,911,800 \text{ m}^3$, $A = 3.8 \cdot 10^6 \text{ m}^2$, $z = 24 \text{ m}$), where FCR = feed conversion ratio, BW = body weight, were used to estimate Nf and SD_N, summarised in Table 2. Seasonal analysis revealed substantial N vulnerability. During the dry season, reduced hydrological flushing caused TN to exceed the oligotrophic threshold, producing strongly negative ΔN values (e.g., -3.515 mg·dm⁻³ in September 2022) and resulting in negative stocking capacities (-841 fish·m⁻²). As with the O₂ model,

Table 2. Seasonal assessment of the vulnerability based on the O₂ and nutrient limits

Date	ΔDO (mg·dm ⁻³)	ΔN (mg·dm ⁻³)	O ₂ limit		N limit		P_r (10 ⁶ kg)	P_l (10 ⁶ kg)	V_u
			SD_{O_2} (fish·m ⁻²)	Bm_{O_2} (10 ⁶ kg)	SD_N (fish·m ⁻²)	Bm_N (10 ⁶ kg)			
Rainy season									
Dec 2022	0.90	−1.41	13	4.329	–	–	4.299	4.299	moderate
Dec 2023	0.00	−0.96	–	–	–	–	–	–	
Dec 2024	0.45	0.16	6	2.149	39	13.249	2.149	–	
Mar 2022	0.40	0.22	6	1.912	52	17.858	1.912	–	
Mar 2023	1.00	0.07	14	4.777	18	6.089	4.777	–	
Mar 2024	−1.10	0.03	–	–	7	2.304	–	–	
Average							2.189	0.716	33%
Dry season									
Jun 2022	0.10	−0.01	1	0.478	–	–	0.478	0.478	moderate
Jun 2023	0.70	0.18	10	3.343	43	14.813	3.343	–	
Jun 2024	−1.20	−0.19	–	–	–	–	–	–	
Sep 2022	0.30	−3.52	4	1.434	–	–	1.434	1.434	
Sep 2023	0.20	0.25	3	0.956	60	20.656	0.956	–	
Sep 2024	−1.20	0.25	–	–	59	20.244	–	–	
Average							1.035	0.319	31%

Explanations: ΔDO = O₂ balances, ΔN = N increment, SD_{O_2} = O₂-based stocking density, Bm_{O_2} = O₂-limited biomass capacities, SD_N = N-based stocking density, Bm_N = N-limited biomass capacities, P_r = production, P_l = production loss, V_u = vulnerability.

Source: own study.

negative SD_N values were truncated to zero, indicating unsustainable conditions rather than biologically meaningful negative densities. In contrast, rainy-season conditions supported positive N buffers, enabling moderate stocking potential (e.g., March 2022: 52 fish·m⁻²; March 2023: 18 fish·m⁻²).

HYPOXIA RISK AND SEASONAL VULNERABILITY

The hazard–exposure–vulnerability (HEV) framework applied follows established environmental risk assessment principles, in which risk is conceptualised as a function of hazard (seasonal hypoxia intensity, ΔDO), exposure (aquaculture biomass and stocking density subject to O₂ and N constraints), and vulnerability (the proportion of production potentially affected by these limitations). The HEV model integrated with O₂⁻ and N-based carrying capacity analyses to evaluate seasonal risks for aquaculture in the Wonorejo Reservoir. Vulnerability ($Vu = \frac{100 P_r}{P_l}$) reflects the proportion of biomass exposed to mortality from O₂ limitation and N stress, classified as very low (0–10%), low (0–20%), moderate (20–40%), high (40–60%), and very high (>60%). Biomass capacity ($Bm = \frac{SD \cdot As}{Z_h}$) was calculated from stocking density (SD), suitable farming area (As), and harvest size (Z_h), while Vu was derived from the ratio of potential production (P_r) to expected production loss (P_l) based on discrepancies between O₂-limited (Bm_{O₂}) and N-limited (Bm_N) biomass capacities.

The combined assessment shows moderate but persistent vulnerability, with Vu values of 31% in the dry season and 33% in the rainy season. During the wet and transitional months (December–March), positive O₂ balances ($\Delta DO = 0.40$ – 1.00 mg·dm⁻³) and moderate N deviations ($\Delta N = -1.41$ to 0.22 mg·dm⁻³) supported sustainable stocking densities of 6–14 fish·m⁻² and Bm_{O₂} values of 1.9 – $4.8 \cdot 10^6$ kg, while high water exchange minimised N exposure, resulting in stable production potential ($Pr = 2.19 \cdot 10^6$ kg vs $P_l = 0.72 \cdot 10^6$ kg).

Risk increased sharply during the dry season (June–September), when negative O₂ balances ($\Delta DO = -1.20$ mg·dm⁻³) and N oversaturation ($\Delta N =$ from -0.19 to -3.52 mg·dm⁻³) eliminated O₂-based carrying capacity ($Bm_{O_2} < 0$) and produced negative production feasibility in several months (e.g., $-5.7 \cdot 10^6$ kg in June 2024). These conditions were driven by minimal inflows, long residence times, weak vertical mixing, and accumulated organic loads, resulting in high vulnerability ($Vu > 40\%$) and confirming O₂ limitation as the dominant ecological constraint, with N loading intensifying long-term instability.

CONCLUSIONS

This study integrates O₂ balance, nitrogen loading, and ecological stability into a seasonal modelling framework to evaluate aquaculture carrying capacity in the Wonorejo Reservoir. Results indicate that, under precautionary ecological thresholds, O₂ availability is the primary modelled constraint on sustainable stocking density, particularly during the dry season, when inflow is reduced and organic respiration increases. Although nutrient levels remained largely within oligotrophic limits, seasonal declines in O₂ surplus constrained effective carrying capacity relative to theoretical nitrogen-based estimates. These findings reflect structural limitations in O₂ buffering capacity rather than observed hypoxic events or fish mortality. The approach relies on

fixed ecological thresholds and a simplified nutrient representation, and therefore it does not capture short-term fluctuations or complex biogeochemical feedbacks. These findings should be interpreted as precautionary, system-level estimates, as reliance on primarily central-station monitoring and limited spatial validation introduces uncertainty in fully representing reservoir-wide heterogeneity. Under these assumptions, the framework offers a decision-support tool to guide seasonal stocking, spatial zoning, and aeration strategies in tropical reservoir aquaculture systems.

SUPPLEMENTARY MATERIAL

https://docs.google.com/document/d/1FCrINYH_y8UP-O259e0wbSIQh7OZrBv/edit#heading=h.3i6t19we21nq

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CONFLICT OF INTERESTS

All authors declare that they have no conflict of interest.

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